

Tutorial 4.

1. a) $y'' + y' - 12y = \sin(2x)$

Solve $y'' + y' - 12y = 0$

$$y = e^{\lambda x} \Rightarrow (\lambda^2 + \lambda - 12)e^{\lambda x} = 0$$

$$(\lambda + 4)(\lambda - 3) = 0 \quad \text{so} \quad \lambda = -4 \text{ or } \lambda = 3$$

$$y_{\text{hom}} = C_1 e^{3x} + C_2 e^{-4x}$$

Try $y_p = K_1 \cos(2x) + K_2 \sin(2x)$

Then $y_p'' + y_p' - 12y_p$

$$= -4y_p + y_p' - 12y_p = y_p' - 16y_p$$

$$= -2K_1 \sin(2x) + 2K_2 \cos(2x) - 16K_1 \cos(2x) - 16K_2 \sin(2x)$$

$$= (\cos 2x)(2K_2 - 16K_1) + (\sin 2x)(-2K_1 - 16K_2)$$

$$= \sin(2x)$$

$$\text{so } K_2 = 8K_1 \quad \text{and} \quad -2K_1 - 16K_2 = 1$$

$$\text{so } -2K_1 - 16(8K_1) = K_1(-2 - 128) = 1$$

$$K_1 = -\frac{1}{130}$$

$$K_2 = -\frac{8}{130} = -\frac{4}{65} \quad \text{so}$$

$$y = C_1 e^{3x} + C_2 e^{-4x} - \frac{\cos(2x)}{130} - \frac{4}{65} \sin(2x)$$

$$b) \quad y'' + 8y' + 16y = 3x^2$$

$$\text{Solve for } y_h \text{ if } y'' + 8y' + 16y = 0$$

$$y = e^{\lambda x} \Rightarrow \lambda^2 + 8\lambda + 16 = 0$$

$$(\lambda + 4)^2 = 0 \quad \Rightarrow \quad \lambda = -4$$

$$\text{so } y_h = C_1 e^{-4x} + C_2 x e^{-4x}$$

$$y_p = ax^2 + bx + c \quad , \quad y_p' = 2ax + b$$

$$y_p'' + 8y_p' + y_p = 2a + 8(2ax + b) + 16(ax^2 + bx + c) = 3x^2$$

$$\text{so } 16a = 3 \text{ and } a = \frac{3}{16}$$

$$\text{and } 16a + 16b = 0 \text{ so } b = -a = -\frac{3}{16}$$

$$\text{and } 2a + 8b + 16c = 0$$

$$\Rightarrow 16c = -2a - 8b = -\frac{6}{16} - 8\left(-\frac{3}{16}\right)$$

$$16c = \frac{18}{16} = \frac{9}{8}$$

$$c = \frac{9}{16 \times 8} = \frac{9}{128}$$

$$\text{so } y = C_1 e^{-4x} + C_2 x e^{-4x} + \frac{3x^2}{16} - \frac{3x}{16} + \frac{9}{128}$$

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1 c) $y'' + 2y' - 8y = 5e^{2x}$

solve for y_h given $y_h'' + 2y_h' - 8y_h = 0$

$$y_h = e^{\lambda x} \Rightarrow (\lambda^2 + 2\lambda - 8)e^{\lambda x} = 0$$

$$(\lambda + 4)(\lambda - 2) = 0 \quad \text{so } \lambda = -4 \quad \text{or } \lambda = 2$$

$$\text{set } y_h = C_1 e^{2x} + C_2 e^{-4x}$$

Try $y_p = K e^{2x}$ (see what goes wrong!).

$$\begin{aligned} \text{Then } y_p'' + 2y_p' - 8y_p &= 4K e^{2x} + 2K e^{2x} - 8K e^{2x} \\ &= -2K e^{2x} = 5e^{2x} \end{aligned}$$

$$\text{so } K = -\frac{5}{2}$$

$$\text{so } y_p = -\frac{5}{2} e^{2x}$$

$$\text{and } y = y_h + y_p = C_1 e^{2x} + C_2 e^{-4x} - \frac{5}{2} e^{2x}$$

↓ This didn't yield the particular soln.

* $\Rightarrow$ use $y_p = Kx e^{2x}$,

$$\text{then } y_p' = 2Kx e^{2x} + K e^{2x}$$

$$\begin{aligned} \text{and } y_p'' &= 4Kx e^{2x} + 2K e^{2x} + 2K e^{2x} \\ &= 4Kx e^{2x} + 4K e^{2x} \end{aligned}$$

$$\text{set } y_p'' + 2y_p' - 8y_p = 5e^{2x}$$

$$\text{Then } 4Kxe^{2x} + 4Ke^{2x} + 2(2Kxe^{2x} + Ke^{2x}) - 8Kxe^{2x} = 5e^{2x}$$

$$\text{so } 4Kx + 4K + 4Kx + 2K(-8Kx) = 5$$

$$6K = 5$$

$$K = \frac{5}{6}$$

$$\text{so } y = y_h + y_p = c_1 e^{2x} + c_2 e^{-4x} + \frac{5}{6} x e^{2x}$$

$$d) \quad y'' + 4y' + 6y = 7$$

$$\text{solve for } y_h \text{ if } y_h'' + 4y_h' + 6y_h = 0$$

$$y_h = e^{\lambda x} \text{ so } (\lambda^2 + 4\lambda + 6)e^{\lambda x} = 0$$

$$\text{so } \lambda^2 + 4\lambda + 6 = 0$$

$$\text{so } \lambda = \frac{-4 \pm \sqrt{16 - 4(6)}}{2} = -2 \pm \frac{\sqrt{-8}}{2}$$

$$\lambda = -2 \pm i\sqrt{2}$$

$$\text{so } y_h = e^{-2x} [A \cos(\sqrt{2}x) + B \sin(\sqrt{2}x)]$$

$$y_p = K \Rightarrow 6K = 7 \text{ or } K = \frac{7}{6}$$

$$\text{so } y = y_h + y_p = e^{-2x} [A \cos(\sqrt{2}x) + B \sin(\sqrt{2}x)] + \frac{7}{6}$$

$$2a) y'' + 4y' + 3y = 3x$$

$$y(0) = 2 \quad \text{Tutoral 4}$$

Solve for y_h given

$$y'(0) = -1$$

$$y_h'' + 4y_h' + 3y_h = 0$$

$$y_h = e^{\lambda x} \Rightarrow (\lambda^2 + 4\lambda + 3)(e^{\lambda x}) = 0$$

$$\Rightarrow \lambda^2 + 4\lambda + 3 = (\lambda + 1)(\lambda + 3) = 0$$

$$\Rightarrow \lambda = -1 \text{ or } \lambda = -3$$

$$y_h = C_1 e^{-x} + C_2 e^{-3x}$$

$$\text{Try } y_p = ax + b, \text{ then } y_p' = a$$

$$\text{Then } y_p'' + 4y_p' + 3y_p = 3x = 0 + 4a + 3(ax + b)$$

$$\text{so } a = 1 \text{ and } 4a + 3b = 0 \quad \text{so } 3b = -4$$

$$b = -\frac{4}{3}$$

$$y_p = x - \frac{4}{3}$$

$$\text{so } y = C_1 e^{-x} + C_2 e^{-3x} + x - \frac{4}{3}$$

$$y(0) = 2 = C_1 + C_2 - \frac{4}{3}$$

$$y'(0) = -1 = -C_1 - 3C_2 + 1$$

$$\text{so } 1 = -2C_2 - \frac{1}{3} \quad \left| \text{ so } C_1 = 2 + \frac{2}{3} + \frac{4}{3} = 4 \right.$$

$$\frac{4}{3} = -2C_2 \quad \left| \begin{array}{l} y = 4e^{-x} - \frac{2}{3}e^{-3x} + x - \frac{4}{3} \\ C_2 = -\frac{2}{3} \end{array} \right.$$

$$2b) \cdot y'' - 4y' + 4y = \cos(x)$$

$$\text{Set } y_h'' - 4y_h' + 4y_h = 0$$

$$y(0) = 3 \text{ and } y'(0) = 0$$

$$\text{Then } y_h = e^{\lambda x}$$

$$\text{so } (\lambda^2 - 4\lambda + 4)e^{\lambda x} = 0$$

$$\text{so } (\lambda - 2)^2 = 0 \text{ so } \lambda = 2$$

$$y_h = c_1 e^{2x} + c_2 x e^{2x} \text{ (repeated roots)}$$

$$y_p = K_1 \cos x + K_2 \sin x, \quad y_p' = -K_1 \sin x + K_2 \cos x$$

$$y_p'' - 4y_p' + 4y_p = -K_1 \cos x - K_2 \sin x$$

$$+ 4K_1 \sin x - 4K_2 \cos x + 4K_1 \cos x + 4K_2 \sin x$$

$$= \cos(x)$$

$$\text{so } \begin{aligned} -K_1 - 4K_2 + 4K_1 &= 1 \\ 3K_1 - 4K_2 &= 1 \end{aligned}$$

$$\text{and } -K_2 + 4K_1 + 4K_2 = 0 \text{ so } 3K_2 = -4K_1$$

$$K_2 = -\frac{4K_1}{3}$$

$$\text{so } 3K_1 - 4\left(-\frac{4K_1}{3}\right) = 1 \quad \frac{3 \times 3K_1 + 16K_1}{3} = 1$$

$$\frac{25K_1}{3} = 1 \text{ so } K_1 = \frac{3}{25} \text{ and } K_2 = -\frac{4}{3} \times \frac{3}{25} = -\frac{4}{25}$$

$$\text{so } y = y_h + y_p = c_1 e^{2x} + c_2 x e^{2x} + \frac{3}{25} \cos x - \frac{4}{25} \sin x$$

2 b) ctd.

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$$y = c_1 e^{2x} + c_2 x e^{2x} + \frac{3}{25} \cos x - \frac{4}{25} \sin x$$

$$y'(x) = 2c_1 e^{2x} + 2c_2 x e^{2x} + c_2 e^{2x} - \frac{3}{25} \sin x - \frac{4}{25} \cos x$$

$$y'(0) = 0 \Rightarrow 2c_1 + c_2 - \frac{4}{25} = 0$$

$$y(0) = 3 \Rightarrow c_1 + \frac{3}{25} = 3$$

$$c_1 = \frac{75}{25} - \frac{3}{25} = \frac{72}{25}$$

$$\text{and } c_2 = \frac{4}{25} - 2c_1 = \frac{4}{25} - \frac{144}{25}$$

$$= \frac{-140}{25} = -\frac{28}{5}$$

$$\text{so } y = \frac{72}{25} e^{2x} - \frac{28}{5} x e^{2x} + \frac{3}{25} \cos x - \frac{4}{25} \sin x$$

Tutorial 4 ctd

$$2c) \quad y'' + 2y' + 6y = e^{3x}$$

Solve for y_h given $y_h'' + 2y_h' + 6y_h = 0$.

$$\text{Try } y_h = e^{\lambda x}, \text{ then } (\lambda^2 + 2\lambda + 6)e^{\lambda x} = 0$$

$$\text{so } \lambda^2 + 2\lambda + 6 = 0$$

$$\text{so } \lambda = \frac{-2 \pm \sqrt{4 - 4(6)}}{2}$$

$$\lambda = -1 \pm i \frac{\sqrt{20}}{2} = -1 \pm \sqrt{5}i$$

$$y_h = e^{-x} (A \sin(\sqrt{5}x) + B \cos(\sqrt{5}x))$$

$$\text{Try } y_p = Ke^{3x}$$

$$\text{Then } y_p' = 3Ke^{3x} \text{ and } y_p'' = 9Ke^{3x}$$

$$\text{so } (9K + 6K + 6K)e^{3x} = e^{3x}$$

$$\text{so } 21K = 1 \text{ and so } K = \frac{1}{21}$$

$$\text{Then } y = y_h + y_p = e^{-x} (A \sin(\sqrt{5}x) + B \cos(\sqrt{5}x)) + \frac{e^{3x}}{21}$$

$$\text{Sub } y(0) = 3 \text{ and } y'(0) = -2$$

$$y' = -e^{-x} (A \sin(\sqrt{5}x) + B \cos(\sqrt{5}x)) + e^{-x} \sqrt{5} (A \cos(\sqrt{5}x) - B \sin(\sqrt{5}x)) + \frac{e^{3x}}{7}$$

$$y'(0) = -(B) + \sqrt{5}A + \frac{1}{7} = -2; y(0) = 3 = B + \frac{1}{21}$$

2c) ctd.

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$$\text{so } B = 3 - \frac{1}{21} = \frac{63}{21} - \frac{1}{21} = \frac{62}{21}$$

$$\text{and } -2 - \frac{1}{7} + B = \sqrt{5}A$$

$$\text{so } \sqrt{5}A = \frac{62}{21} - \frac{3}{21} - \frac{42}{21} = \frac{17}{21}$$

$$A = \frac{17}{21\sqrt{5}} = \frac{17 \times \sqrt{5}}{105}$$

$$\text{so } y = e^{-x} \left(\frac{17\sqrt{5}}{105} \sin(\sqrt{5}x) + \frac{62}{21} \cos(\sqrt{5}x) \right) + \frac{e^{3x}}{7}$$

$$H. \quad my'' + cy' + ky = 5 \sin(\omega t) \quad \begin{matrix} m=1, c=0 \\ k=9 \end{matrix}$$

$$y'' + 9y = 5 \sin(\omega t)$$

$$\text{solve } y'' + 9y = 0 \quad \text{Try } y(t) = e^{\lambda t}$$

$$(\lambda^2 + 9) = 0 \quad \lambda^2 = -9 \quad \text{so } \lambda = \pm 3i$$

no decay here as $c=0$.

damping $\Rightarrow$ decay

no damping $\Rightarrow$ no decay.

$\omega_0 = 3$ here.

$5 \sin(\omega t)$
replace by $5 \cos(\omega t)$
to co-incide

$$y_h = A \cos(3t) + B \sin(3t) \quad \left. \begin{matrix} \text{with lecture} \\ \text{notes} \end{matrix} \right\}$$

$$y_p = a \cos(\omega t) + b \sin(\omega t)$$

$$y_p' = -a\omega \sin(\omega t) + b\omega \cos(\omega t)$$

$$y_p'' = -a\omega^2 \cos(\omega t) - b\omega^2 \sin(\omega t)$$

$$= -\omega^2 y_p$$

Now $y'' + 9y = 5 \sin(\omega t)$ } H dtd with
 so $-w^2 y_p + 9y_p = 5 \sin(\omega t)$ } wrong for ω
 fn

$$(a \cos(\omega t) + b \sin(\omega t))(9 - w^2) = 5 \sin(\omega t)$$

so $(9 - w^2)a = 0$ and $(9 - w^2)b = 5$

so $a = 0$ and $b = \frac{5}{9 - w^2} = \frac{5}{w_0^2 - w^2}$

so $y_p = \left(\frac{5}{9 - w^2} \right) \sin(\omega t) = \frac{F_0}{m(\omega_0^2 - \omega^2)}$ since $F_0 = 5$ & $m = 1$.

so $y = A \cos(3t) + B \sin(3t) + \frac{5}{(9 - w^2)} \sin(\omega t)$

$w_0 = 3$ here.

$y(0) = A = 0$

$y'(t) = -3A \sin(3t) + 3B \cos(3t)$

$+ \frac{5}{(9 - w^2)} w \cos(\omega t)$

$y'(0) = 0 + 3B + \frac{5w}{9 - w^2}$

$3B = - \frac{5w}{w_0^2 - w^2} \quad B = \frac{-5(w/w_0)}{w_0^2 - w^2}$

$y = \frac{-5(w/w_0)}{w_0^2 - w^2} \sin(\omega t) + \frac{5}{(w_0^2 - w^2)} \sin(\omega t)$
 $y(0) = 0$ and $y'(0) = 0$ and $w_0 = 3$

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$$3.a) my'' + cy' + ky = 3 \cos(3t) \quad \begin{array}{l} m=1 \\ c=4 \\ k=3 \end{array}$$

$$\text{set } my_h'' + cy_h' + ky_h = 0$$

$$\text{try } y_h = e^{\lambda t} \text{ then } \lambda^2 + \frac{c}{m}\lambda + \frac{k}{m} = 0$$

$$\text{or } m\lambda^2 + c\lambda + k = 0$$

$$\text{so } \lambda = \frac{-c \pm \sqrt{c^2 - 4km}}{2m} \quad \begin{array}{l} \text{here } c^2 - 4km \\ = 16 - 4(3)(1) \\ = 4 > 0 \end{array}$$

$$\lambda = \frac{-4 \pm 2}{2} = -2 \pm 1 \quad \text{so } \lambda = -3 \text{ or } -1$$

$$y_h = c_1 e^{-t} + c_2 e^{-3t}$$

$$\text{now let } y_p = K_1 \cos(3t) + K_2 \sin(3t)$$

$$\text{Then } m(-9)y_p + c(-3K_1 \sin(3t) + 3K_2 \cos(3t)) + ky_p = 3 \cos(3t)$$

$$(k - 9m)(K_1 \cos(3t) + K_2 \sin(3t)) - 3K_1 c \sin(3t) + 3K_2 c \cos(3t) = 3 \cos(3t)$$

equate coefficients:

$$(k - 9m)K_1 + 3K_2 c = 3$$

$$\begin{aligned} \text{so } (3 - 9)(K_1) + 12K_2 &= 3 \\ -6K_1 + 12K_2 &= 3 \\ -2K_1 + 4K_2 &= 1 \quad (1) \end{aligned}$$

$$\text{and } (k - 9m)(K_2) = 3K_1 c$$

$$-6K_2 = 12K_1$$

$$K_2 = -2K_1 \quad (2)$$

$$\begin{aligned} \text{so } -2K_1 + 4(-2K_1) &= 1 \\ -10K_1 &= 1, K_1 = -\frac{1}{10} \end{aligned}$$

$$K_2 = +\frac{1}{5}$$

3a) ctd so $y = y_h + y_p$

$$= C_1 e^{-t} + C_2 e^{-3t} - \frac{1}{10} \cos(3t) + \frac{\sin(3t)}{5}$$

b) $m=1, c=0, k=4, r(t)=5\cos(2t)$

$$m y_h'' + c y_h' + k y_h = 0$$

set $y_h = e^{\lambda t}$, find $\lambda = \frac{-c \pm \sqrt{c^2 - 4km}}{2m}$

$$\lambda = \frac{\pm \sqrt{-4(4)(1)}}{2} = \pm \frac{4i}{2} = \pm 2i$$

so $y_h = C_1 \cos(2t) + C_2 \sin(2t)$

No damping, simple harmonic oscillation

$y_p = t [a \cos(2t) + b \sin(2t)]$
 $\uparrow$ if no t , get same solution as before

$$y'' + 4y = 5\cos(2t)$$

$$y_p = at \cos(2t) + bt \sin(2t)$$

$$y_p' = a \cos(2t) + 2at \sin(2t) + b \sin(2t) + 2bt \cos(2t)$$

$$y_p'' = -2a \sin(2t) - 2a \sin(2t) + 4(2at) \cos(2t) + 2b \cos(2t) + 2b \cos(2t) - 4bt \sin(2t)$$

and $y_p'' + 4y_p = 5\cos(2t)$

so $-2a - 2a - 4bt + 4bt = 0 \Rightarrow a = 0$
 and $4b = 5$ so $b = 5/4$

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3b).

$$y = y_A + y_P$$

$$= c_1 \cos(2t) + c_2 \sin(2t) + \frac{5t \sin(2t)}{4}$$

checking: $y_P'' + 4y_P = 5 \cos(2t)$

$\sin(2t): -2a - 2a - 4bt + 4bt = 0$

$\cos(2t): 8at + 2b + 2b + at = 5$

$$a=0 \quad \text{so} \quad 4b=5 \quad \left| \begin{array}{l} \text{Don't evaluate} \\ y'' + 4y' \text{ by mistake} \end{array} \right.$$

In general, get resonance $\omega = \omega_0$

$$y'' + \omega_0^2 y = \frac{F_0}{m} \cos(\omega_0 t)$$

$$\rightarrow y_P = t(a \cos(\omega_0 t) + b \sin(\omega_0 t))$$

$$y_P' = a \cos(\omega_0 t) + b \sin(\omega_0 t) + -\omega_0 t a \sin(\omega_0 t) + \omega_0 b t \cos(\omega_0 t)$$

$$y_P'' = -a \omega_0 \sin(\omega_0 t) + \omega_0 b \cos(\omega_0 t)$$

$$- \omega_0^2 t a \cos(\omega_0 t) - \omega_0 a \sin(\omega_0 t)$$

$$- \omega_0^2 b t \sin(\omega_0 t) + \omega_0 b \cos(\omega_0 t)$$

Since $y_P'' + \omega_0^2 y_P = \frac{F_0}{m} \cos(\omega_0 t)$

$\sin: -2a \omega_0 - \omega_0^2 b t + \omega_0^2 b = 0 \quad \text{so} \quad a = b$

$\cos: \omega_0 b - \omega_0^2 t a + \omega_0 b + 4at = 2\omega_0 b = \frac{F_0}{m}$
 $b = \frac{F_0}{2\omega_0 m}$

H. $y'' + 9y = 5 \cos(\omega t)$

a) $y_h'' + 9y_h = 0$, $\lambda^2 + 9 = 0$, $\lambda = \pm 3i$
 $\omega_0 = 3$

$$y = C_1 \cos 3t + C_2 \sin 3t$$

Assume $\omega \neq 3$?

Then we don't have resonance.

$$y_p = a \cos(\omega t) + b \sin(\omega t)$$

$$y_p' = -a\omega \sin(\omega t) + \omega b \cos(\omega t)$$

$$y_p'' = -a\omega^2 \cos(\omega t) - \omega^2 b \sin(\omega t)$$

$$= -\omega^2 y_p$$

$$y_p'' + 9y_p = 5 \cos(\omega t)$$

$$(9 - \omega^2)(a \cos(\omega t) + b \sin(\omega t)) = 5 \cos(\omega t)$$

$$b = 0 \text{ and } a(9 - \omega^2) = 5$$

$$a = \frac{5}{9 - \omega^2} = \frac{5}{\omega_0^2 - \omega^2}$$

so $y = C_1 \cos(3t) + C_2 \sin(3t) + \frac{5}{\omega_0^2 - \omega^2} \cos(\omega t)$

$\omega \neq 3$

Say $C_1 = -\frac{F_0}{m(\omega_0^2 - \omega^2)} = \frac{-5}{(\omega_0^2 - \omega^2)}$ and

$$C_2 = 0$$

Then $y = \frac{-5}{(\omega_0^2 - \omega^2)} \cos(\omega_0 t) + \frac{5}{\omega_0^2 - \omega^2} \cos(\omega t)$

4a) general solution.

This would fit with $y(0) = 0$ and $y'(0) = 0$. $\omega \neq \omega_0$.

4b) Beats: $\cos(\omega t) - \cos(\omega_0 t)$ ω_0 close to ω

$$= \left(\sin\left(\frac{\omega_0 + \omega}{2} t\right) \right) \left(\sin\left(\frac{\omega_0 - \omega}{2} t\right) \right)$$

$\swarrow$
 $\frac{\omega_0 + \omega}{2}$ close to ω_0 or ω

$\frac{\omega_0 - \omega}{2}$ close to zero, giving long beat period.

2 beats to 2π so $2T \left(\frac{\omega_0 - \omega}{2} \right) = 2\pi$ beat period

$$T(\omega_0 - \omega) = 2\pi$$

$$3 - \omega = \frac{2\pi}{5}$$

$$3 - \frac{2\pi}{5} = \omega$$

frequency gives

5 sec. between beats.

$$\omega = 1.74 \text{ Hz}$$

5. a) $3y'' + y' + 5y = 3\sin(\omega t)$
 $\omega_0^2 = \frac{k}{m} = \frac{5}{3}$ $= 3\cos(\omega t - \frac{\pi}{2})$
 $c=1$ $c^* = 3$
Damped.

Practical resonance. No run-away resonance.

$$\omega_{\max}^2 = \omega_0^2 - \frac{c^2}{2m^2} = \frac{5}{3} - \frac{1}{2 \times 9} = \frac{29}{18}$$

$$\omega_{\max} = 1.27 \text{ Hz}$$

$$c^*(\omega_{\max}) = \frac{2mF_0}{c \sqrt{4m^2\omega_0^2 - c^2}} = \frac{2(3)(3)}{(1) \sqrt{4(9)\frac{5}{3} - 1}}$$

$$= \frac{18}{\sqrt{59}} = 2.34$$

b) $m=1, c=2, k=3, f(t) = \cos(\omega t) \quad F_0=1$

$$\omega_{\max}^2 = \frac{k}{m} - \frac{c^2}{2m^2} = 3 - \frac{4}{2} = 1$$

$$c^*(\omega_{\max}) = \frac{2mF_0}{c \sqrt{4m^2\omega_0^2 - c^2}} = \frac{2(1)(1)}{2 \sqrt{4 \times 3 - 4}}$$

$$= \frac{1}{\sqrt{8}} = 0.35$$

6 a) $m=2, c=3, k=3, r(t)=3\sin(3t)$

$\omega_0 = \frac{3}{2}$, we have damping

$c^2 - 4mk = 9 - 4(2)(3) < 0$

oscillation and damping $\Rightarrow$ damping dies away

$\omega=3, (r(t)=3\sin(3t))$

- transient

so $C^*(\omega) = \frac{F_0}{\sqrt{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}}$ don't forget!

$\omega_0 = \frac{k}{m}$

$3\sin(3t) = 3\cos(3t - \frac{\pi}{2})$

$= \frac{3}{\sqrt{4(\frac{3}{2} - 3)^2 + 9 \times 9}}$

$= \frac{3}{\sqrt{1306}}$ $\mu = \frac{\pi}{2}$

$= 0.1715$

(not 0.175)

b) $m=5, c=2, k=4, r(t)=2\sin(\pi t)$

$C^*(\pi) = \frac{\cancel{F_0} 2}{\sqrt{25(\frac{4}{5} - \pi^2)^2 + \pi^2 \times 4}}$

$= \frac{2}{\sqrt{2056.4431 + \pi^2 \times 4}}$

$= \frac{2}{\sqrt{2055.922}} = 0.0437$

